Pritam

Portfolio: prtmxio

+91-9812015241 | dhimanpritam1579@gmail.com | LinkedIn | GitHub

Technical Skills

Programming Languages: C/C++, Python, Java, SQL, VHDL
Technologies & Frameworks: Spring Boot, Java Swing, AWT, OpenCV, REST APIs, Multithreading
Machine Learning: Linear Regression, Logistic Regression, Random Forest, Scikit-learn
Tools & Platforms: Git, GitHub, Linux, IntelliJ IDEA, VS Code, Vim
Specializations: Artificial Intelligence, Machine Learning, Robotics, Embedded Systems, PCB Design

Education

Deenbandhu Chhotu Ram University of Science and Technology — Sonipat, Haryana
Bachelor of Technology in Electronics & Communication Engineering; CGPA: 8.5/10 — *Sep 2023 - Present*

Central Board of Secondary Education (CBSE) — 2022
Higher Secondary Certificate - Mathematics, Physics, Chemistry; Grade: 96.2% — *New Delhi*

Experience

Embedded Systems Application Developer Intern — Jul 2025 – Present
PCB CUPID, Bangalore, India — *Remote*

- Developed embedded applications and IoT solutions using Glyph development boards (C2, C6, H2) with microcontroller programming, achieving precision control and hardware-software integration for production-ready systems
- Designed and prototyped embedded solutions using C/C++ programming, converting proof-of-concepts to scalable systems deployed in real-world industrial environments
- Implemented microcontroller firmware solutions with sensor integration, achieving seamless communication between hardware components and software algorithms for automated systems

Robotics Engineering Team Member and Technical Mentor — Oct 2023 – Present
THINKBOTS Robotics Club, DCRUST — *Murthal, Haryana*

- Engineered autonomous robotics systems including line-following robots and multi-terrain gripper mechanisms using Arduino microcontrollers, C++ programming, and sensor arrays, achieving 95% accuracy in path tracking algorithms
- Mentored 15+ junior developers on robotics fundamentals, Arduino programming, embedded systems design, and project management, improving team project completion rate by 40% through technical training programs
- Optimized robotics control systems through collaborative debugging, algorithm enhancement, and system integration, reducing development and troubleshooting time by 30% across multiple team projects

Projects

XTorr - Multi-Threaded BitTorrent P2P Client — GitHub — *Java, Networking, Concurrency* — Dec 2024

- Developed complete peer-to-peer BitTorrent client from scratch using Java, implementing multi-threaded architecture, TCP/IP socket programming, peer-wire protocol communication, tracker HTTP requests, and SHA-1 cryptographic validation for reliable file sharing
- Engineered high-performance concurrent download system using thread pools, ExecutorService, and synchronized data structures to fetch file pieces simultaneously from multiple peers, achieving 300% performance improvement over sequential downloads

Forge - PCB Design Visualization CLI Application — GitHub — *Python, SVG Processing* — Aug 2025

- Created command-line interface application that automates conversion of KiCad SVG files into comprehensive PCB visualizations, implementing geometric mapping algorithms, coordinate transformation systems, and component placement automation with sub-millimeter precision
- Developed modular SVG processing pipeline with XML parsing, file I/O operations, and mathematical coordinate transformation algorithms, streamlining PCB design workflow for electrical engineers and reducing manual design time by 60%

Autonomous Navigation Line Following Robot — *Arduino Nano, C++, PID Control* Mar 2025

- Built autonomous mobile robot with 8-IR sensor array, L298N motor driver, and custom PID control algorithm implementation, achieving 98% path accuracy through complex navigation scenarios including sharp corners and varying track conditions
- Optimized sensor calibration protocols, implemented digital filtering algorithms, and developed adaptive thresholding systems, reducing false positive readings by 85% and ensuring stable performance across diverse lighting conditions and surface materials

Dobosaurus - Interactive Game Development Project — GitHub — *Java, Swing GUI* Nov 2024

- Developed arcade-style platformer game application using Java Swing framework, implementing responsive user input handling, procedural obstacle generation algorithms, collision detection systems, and real-time scoring mechanisms optimized for 60+ FPS performance
- Designed scalable object-oriented software architecture with separation of concerns, implementing MVC design patterns to separate game logic, graphics rendering, and user input systems for maintainable and extensible codebase

Achievements & Recognition

Smart India Hackathon - National Level Competition — *Team Leader & Software Developer* 2024

- Led development of AI-powered Face Recognition Surveillance System and Communication Platform for Missing Persons tracking at Simhastha Ujjain, implementing computer vision algorithms, database management, and real-time notification systems for Police Department

Additional Skills

Operating Systems: Linux (Ubuntu, Arch Linux, CentOS), Windows, Unix
Development Tools: Git, GitHub, Docker, Jenkins, Vim, Make, CMake, Debugging Tools
Soft Skills: Team Leadership, Project Management, Technical Documentation, Problem Solving, Agile Development